

AIM

To write a C program using structures and functions to store the details of books such as Book ID, Book Name, and Price, upload details of five books, and display the information of a book based on the given Book ID.

PROCEDURE

1. Include the required header file `stdio.h`.
2. Define a structure `Book` with members `id`, `name`, and `price`.
3. Declare function to Upload book details, Display book details based on Book ID.
4. In the `main()` function:
 - Declare an array of structures to store details of five books.
 - Call the function to upload book details.
 - Get the Book ID from the user to search.
 - Call the function to display book details.
5. Compile and execute the program.

CODE

/*

Name of the File : `structure.c`

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Date : 17-01-2026

Purpose : To develop a simple Library Management System
using structures and functions in C to store
and display book details based on Book ID.

```
*/#include <stdio.h>
```

```
#define SIZE 5
```

```
struct Book
```

```
{
```

```
    int id;
```

```
    char name[50];
```

```
    float price;
```

```
};
```

```
void UploadBookDetails(struct Book b[]);
```

```
void PrintBookByID(struct Book b[], int searchID);
```

```

int main()
{
    struct Book books[SIZE];

    int searchID;

    UploadBookDetails(books);

    printf("\nEnter Book ID to search: ");

    scanf("%d", &searchID);

    PrintBookByID(books, searchID);

    return 0;
}

void UploadBookDetails(struct Book b[])
{
    int i;

    for (i = 0; i < SIZE; i++)
    {
        printf("\nEnter details for Book %d\n", i + 1);

        printf("Book ID: ");

        scanf("%d", &b[i].id);

        printf("Book Name: ");

        scanf(" %[^\\n]", b[i].name);

        printf("Book Price: ");

        scanf("%f", &b[i].price);

    }
}

void PrintBookByID(struct Book b[], int searchID)
{
    int i, found = 0;

    for (i = 0; i < SIZE; i++)
    {
        if (b[i].id == searchID)
        {
            printf("\nBook Found");
        }
    }
}

```

```

        printf("\nBook ID : %d", b[i].id);

        printf("\nBook Name : %s", b[i].name);

        printf("\nBook Price: %.2f\n", b[i].price);

        found = 1;

        break;

    }

}

if (found == 0)

{   printf("\nBook with ID %d not found.\n", searchID); }

}

```

OUTPUT

```

PROBLEMS  TERMINAL  OUTPUT  PORTS
● PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week2_practice3\" ; if ($?)
{ gcc structure.c -o structure } ; if ($?) { .\structure }

Enter details for Book 1
Book ID: 1
Book Name: maths
Book Price: 50

Enter details for Book 2
Book ID: 2
Book Name: computer
Book Price: 100

Enter details for Book 3
Book ID: 3
Book Name: english
Book Price: 70

Enter details for Book 4
Book ID: 4
Book Name: IOT
Book Price: 150

Enter details for Book 5
Book ID: 5
Book Name: OS
Book Price: 120

Enter Book ID to search: 4

Book Found
Book ID : 4
Book Name : IOT
Book Price: 150.00

```

RESULT

Thus, the C program to implement a Library Management System using structures and functions was successfully executed. The program correctly stored the details of five books and displayed the book information based on the given Book ID.